

Results for ChIPSeq Pipeline (v0.9.0)

Output folders:

metrics: PICARD metrics from mapping phase

alignments: RAW unprocessed BAM files. Note before using the BAM's we post process them according to the ENCODE recipes which does the following:

- remove unmapped, mate unmapped, non-proper paired reads
- remove reads with MAPQ < 30
- remove duplicates
- remove Failed QC reads

chipSeq/macs: output of running macs2 callpeak on BED files derived from the post processed bams. The peak type is chosen once for the whole run, not per sample. Broad peaks are the default, for IP's with non-focal/broad binding:

- --broad

for IP's with focal transcription factors the pipeline is run with -n, which calls the narrow peak set instead (no --broad).

In both cases the following arguments are used:

- -p 0.01 - peak cutoff on the p-value rather than the q-value
- --nomodel --shift 0 --extsize <FRAGLEN> - skip MACS model building and extend reads by a fixed fragment length. <FRAGLEN> is the median of the per-sample fragment lengths estimated by macs2 predictd, so a single value is used for every sample in the run.

See the MACS2 website [<https://github.com/taoliu/MACS>] for more information on the output.

There is also an aggregate peak file there which combines the peaks from all samples - macs/macs-PeaksMerged.saf and a file that counts the coverage for each sample within these combined peaks - macs/peaks_raw_fcCounts.txt

chipSeq/qc: QC report/plots for ChIP related QC.

- qcChIPSeq_PROJECT-NUM_.xlsx - is a report for the total number of peaks found and the number of significant peaks. Low number of peaks could indicate an issue with the ChIP.
- qcChIPSeq_PROJECT-NUM_.pdf - plots of the number/percentage of mapped reads that fall in MACS peaks and a PCA plot of the aggregate peak counts for the samples.

chipSeq/bw: Normalized (to 10million reads) bigwig files for loading IP profiles into IGV [<http://software.broadinstitute.org/software/igv/>].

chipSeq/annotate: Annotation of MACS2 peak file using HOMER annotatePeaks.pl. See the HOMER website [<http://homer.ucsd.edu/homer/ngs/annotation.html>] for more details on the output.

chipSeq/diff: Differential analysis of the aggregate peak counts using edgeR, run for each requested pairwise group comparison. Only present if a differential analysis was requested.

- *_DiffPeaksEdgeR_V3.xlsx - differential peaks per comparison, with significant peaks annotated using CHIPseeker. Significance cutoff is FDR < 0.05.
- *_DiffPeaks_V3.pdf - PCA plot plus MA and volcano plots for each comparison.

Comparisons are specified as group pairs; the sign convention is Group2 - Group1, so a positive logFC means higher in Group2.